



**FIRST SEMESTER 2026-2027
COURSE HANDOUT PART II**

Date: 31/07/2026

In addition to part I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course Code : ME F321

Name of the course : Data Mining in Mechanical Sciences

Instructor-in-charge : Madhurjya Dev Choudhury

I. Course Description:

Introduction to data mining, clustering, classification & association, sequence analysis, regressions method, decision trees, machine learning, neural networks, SVM for engineering and manufacturing applications, data mining for product design, applications of data mining in production, planning and scheduling, data mining for defect identification, process and quality control, application of data mining in maintenance, data analysis for machine and manufacturing process monitoring, introduction to cyber physical systems, introduction to big data handling and big data analytics.

II. Scope and Objective of the course:

The primary focus of this course is on application of data mining methods in mechanical sciences and engineering. Following are the details objectives of this course:

- ☐ Understanding the fundamental of data mining.
- ☐ Building mathematical foundation for data mining.
- ☐ Knowledge of data capturing to minimize the information losses.
- ☐ Understanding of patterns in data, data classification and clustering.
- ☐ Knowledge of application of identified patterns in data in fields like in-processing monitoring, machine monitoring, defect analysis, machine and maintenance scheduling etc.
- ☐ Understanding of different prediction methods based on the data processing.
- ☐ Understand the difference between artificial intelligence, machine learning and deep learning.
- ☐ Knowledge of which data mining methods should be used for what type of data.
- ☐ Understanding the concept of big data in mechanical and manufacturing domain and how to handle it.

III. Textbook:

- T1. Kantardzic, Mehmed. Data mining: concepts, models, methods, and algorithms. John Wiley & Sons, 2011.
- T2. Han, J., Kamber, M., and Pei, J. Data mining: Concepts and Techniques. 3rd Edition, Morgan Kaufmann Publishers, July 2011.

IV. Reference Book:

- R1. P.-N. Tang, M. Steinback, and V. Kumar: Introduction to Data Mining, Addison Wesley, Second Edition.





V. Course Contents

L No.	Learning Objectives	Topic to be covered	References
1-2	Understanding the objectives of data mining	Why data analysis? Importance of data analysis, Data analysis to data mining, Definition of data mining, Objectives of data mining, Analogy of data mining with concept of mechanical science, Steps in data mining/Data mining process, Concept of data warehouses, Why data mining fails?	Class notes and T1(1)
3-4	Measuring the data and changing the form of raw data (Data preprocessing)	Data measurement (Sampling method) Structure/representation of data Temporal and spatial data Different types of data; Numeric and categorical data Dimensionality of data, Mean, variance, data normalization methods, data smoothing methods.	Class notes and T2(2,3)
5-6	Statistical analysis and outlier detection	Point estimation, central limit theory, Regression methods, logistical regression, Density and proximity-based outlier detection, clustering based method.	Class notes and T1(2, 5)
7-10	Feature identification/extraction methods	Supervised and unsupervised feature, Architecture for feature subset selection, Principal component analysis for feature selection, relief algorithm, ranking score, entropy method, ChiMerge technique for feature discretization	Class notes and T1(3), T2(3)
11-15	Different classification algorithms and its application	General approach for classification problem, accuracy and error rate, Decision tree, design of decision tree, algorithm for decision tree, K nearest neighbors, support vector machine algorithm	Class notes and T1(4, 6), T(8)
16-20	Different clustering algorithms and its applications	Objective of clustering, methods for similarity measure, Agglomerative hierarchical clustering, partitional and incremental clustering, DBSCAN algorithm, clustering validity.	Class notes and T1(9), T2(10)
21-26	Prediction from data; machine learning methods	Learning machine, types of learning methods; supervised and unsupervised learning, neural networks, autoencoders	Class notes and T1(4,7), R1(4)
27-28	Case studies	Different case studies for application of machine learning in mechanical system, design and manufacturing.	Class notes, Journal papers

Tentative List of Coding Tasks

Week	Experiment No.	Experiment details
1	E1	Basics of python, installing python and setting up python environment; how python works?
2	E2	Getting started with python programming; running code, input, processing and output, running a script





3	E3	Data loading and preprocessing with pandas, data processing with NumPy, Creating NumPy arrays
4	E4	Writing control loop, programming for if and if-else statements, Design and functions
5	E5	Binary classification and multilabel classification, linear regression, logistical regression
6	E6	The data pipeline, principle component analysis Kernel PCA, outlier detection binary classification
7	E7	Feature selection, univariate selection, recursive elimination
8	E8	K-nearest neighbors, Clustering using python DBSCAN in python
9	E9	Programming for decision tree and entropy analysis, different examples
10	E10	SVM for classification, SVM for regression, tuning SVM
11	E11	Neural network programming
12	E12	ANN and deep learning using Keras, different case studies using Keras.

VI. Evaluation Scheme and Schedule:

Evaluation Components	Duration	Weightage (%)	Date and time	Remarks
Mid-semester Test	90 min	20	<TEST_1>	CB/OB
Coding		20	To be announced	
Project/Assignment		20	To be announced	
Quiz		10		
Comprehensive Examination	3 Hours	30	<TEST_C>	CB/OB
TOTAL				

VII. Chamber Consultation Hour: Tuesday 2pm-3pm in 1216-I.

VIII. Notices concerning the course: All notices concerning the course will be communicated via email.

IX. Make-up Policy: Make up will be permitted only in genuine cases with prior permission. Other than medical reason prior permission of IC is compulsory. No make up for quizzes.

X. Minimum Passing Marks/NC: The passing marks for this course will be determined based on one of the two criteria: 30% of the median of the post-comprehensive examination score OR 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as passing marks in this course. NC will be awarded if a student does not appear in the mid-semester or comprehensive examination.

Instructor-in-Charge

ME F321

